

Supplementary Material: Beyond Replay RMSE

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1 Condition-level observer selection

Table 1 reports the closed-loop oracle and the observers chosen by replay position RMSE and LCSE. Regret is the increase in closed-loop cross-track RMSE relative to the oracle. The corresponding regret matrix is shown in Figure 1.

Table 1: Condition-level selection. Regret is in centimetres.

Condition	CL oracle	Pose choice	Pose regret	LCSE choice	LCSE regret
Blackout 4	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Blackout 10	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Blackout 16	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Blackout 22	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Blackout 28	EKF	SSM-EKF	2.35	EKF	0.00
Blackout 34	EKF	SSM-EKF	2.93	EKF	0.00
Range noise 0.05	EKF	EKF	0.00	GRU-EKF	1.39
Range noise 0.12	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Range noise 0.2	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Range noise 0.3	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Range noise 0.4	GRU-EKF	SSM-EKF	2.07	GRU-EKF	0.00
Range noise 0.5	GRU-EKF	SSM-EKF	3.47	GRU-EKF	0.00
Landmarks 2	SSM-EKF	SSM-EKF	0.00	GRU-EKF	9.40
Landmarks 3	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Landmarks 4	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Landmarks 5	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Landmarks 6	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Landmarks 8	SSM-EKF	GRU-EKF	1.40	GRU-EKF	1.40
Gyro bias 0.0	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Gyro bias 0.05	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Gyro bias 0.1	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Gyro bias 0.15	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Gyro bias 0.2	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00
Gyro bias 0.25	GRU-EKF	GRU-EKF	0.00	GRU-EKF	0.00

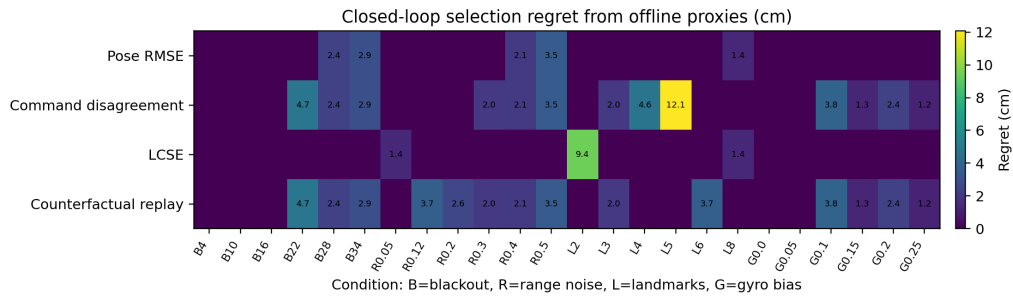


Figure 1: Closed-loop selection regret for each offline proxy and condition. B denotes blackout length, R range noise, L landmark count, and G gyro bias.

2 Matched ablations and training initialization

The paired intervals in Table 2 are computed over matched environment seeds and report the difference relative to the selective+mask model. Negative values favor the ablation. Figure 2 visualizes the same-anchor results and the three independently trained SSM-EKF checkpoints.

Table 2: Full matched EKF-anchor ablation results. Values are mean $\pm$ 95% CI half-width; difference intervals are paired ablation-minus-selective intervals.

Condition	Selective	Non-selective	Paired Δ CI	No global mask	Paired Δ CI
Nominal	0.199 $\pm$ 0.034	0.203 $\pm$ 0.034	[-0.051,0.059]	0.195 $\pm$ 0.028	[-0.044,0.031]
Long blackout	0.336 $\pm$ 0.070	0.235 $\pm$ 0.027	[-0.182,-0.020]	0.308 $\pm$ 0.060	[-0.102,0.040]
High range noise	0.307 $\pm$ 0.090	0.322 $\pm$ 0.043	[-0.093,0.103]	0.294 $\pm$ 0.044	[-0.110,0.067]
Two landmarks	0.289 $\pm$ 0.050	0.235 $\pm$ 0.024	[-0.103,-0.007]	0.274 $\pm$ 0.038	[-0.050,0.018]
High gyro bias	0.314 $\pm$ 0.065	0.393 $\pm$ 0.039	[0.011,0.145]	0.312 $\pm$ 0.039	[-0.055,0.051]

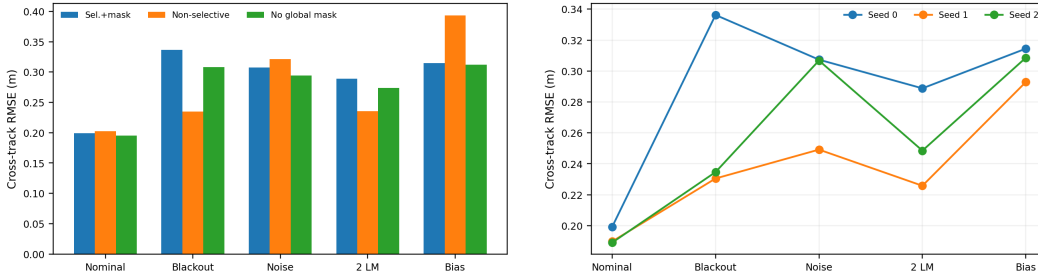


Figure 2: Same-anchor ablation (left) and training-initialization sensitivity (right).

Table 3: Across-training-seed SSM-EKF summary.

Condition	Minimum mean	Maximum mean	Across-seed SD
Nominal	0.189	0.199	0.006
Long blackout	0.231	0.336	0.060
High range noise	0.249	0.307	0.033
Two landmarks	0.226	0.289	0.032
High gyro bias	0.293	0.314	0.011

3 Controller sensitivity

The five-seed Kanayama sensitivity is not used as the primary statistical result. It tests whether conclusions transfer when the same observers, plant, actuator limits, conditions, and random seeds are paired with a different tracking law.

4 Full MRCLAM EKF grid and calibrated replay

The best position result is obtained with process scale $q = 0.25$ and measurement scale $r = 4$. In the separate real-noise-calibrated simulated replay, dead reckoning is the closed-loop oracle. Pose RMSE selects it with zero regret, whereas command disagreement chooses EKF $q = 1, r = 4$ with 0.120 m regret and LCSE chooses EKF $q = 0.25, r = 4$ with 0.203 m regret.

5 Reproducibility map

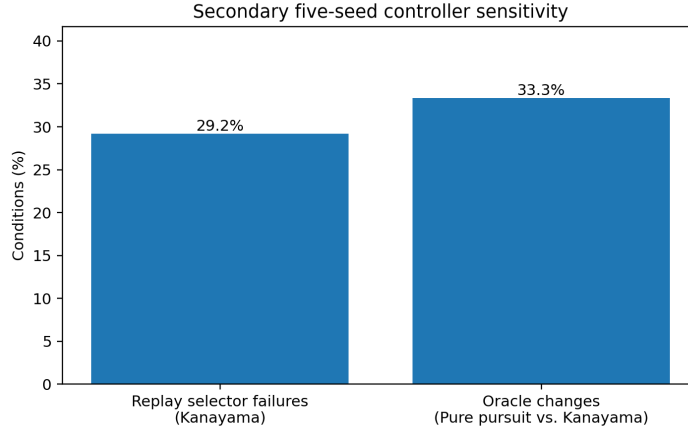


Figure 3: Replay-RMSE selection failures for Kanayama and the fraction of conditions in which the pure-pursuit and Kanayama closed-loop oracles differ.

Statistic	Result
Replay-RMSE failures for Kanayama	7/24 (29.2%)
Mean / maximum Kanayama regret	0.0142 / 0.0923 m
Closed-loop oracle changes	8/24 (33.3%)
Oracle-change bootstrap CI	0.208–0.625

Table 4: Physical-log localization on the 15-landmark MRCLAM segment.

Observer / covariance scale	Pos. RMSE	Head. RMSE	Command diff.	LCSE
DR	3.392	1.662	0.586	62.91
EKF q0.25 r0.25	1.599	0.864	0.501	38.13
EKF q0.25 r1	1.511	0.879	0.508	32.10
EKF q0.25 r4	1.357	0.850	0.525	4.22
EKF q1 r0.25	1.703	0.879	0.489	39.58
EKF q1 r1	1.599	0.864	0.501	38.13
EKF q1 r4	1.507	0.878	0.507	32.10
EKF q4 r0.25	1.804	0.881	0.489	36.31
EKF q4 r1	1.702	0.879	0.489	39.58
EKF q4 r4	1.597	0.863	0.500	38.13

Table 5: Primary commands in the released package.

Task	Command
Train checkpoints	<code>python scripts/run_train.py -backend torch -force</code>
Synthetic evaluation	<code>python scripts/run_eval.py</code>
Replay/closed-loop proxy audit	<code>python scripts/run_proxy_analysis.py</code>
Matched EKF ablations	<code>python scripts/run_matched_ablation.py</code>
Training-seed study	<code>python scripts/run_training_seed_study.py</code>
Second controller	<code>python scripts/run_second_controller.py -seeds 5</code>
Download MRCLAM	<code>python scripts/download_mrclam.py</code>
Physical-log analysis	<code>python scripts/run_mrclam.py</code>
Publication figures	<code>python scripts/make_enhanced_figures.py</code>